The Geography Of Interest Rate Differences

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Abstract

This paper studies how geography shapes international capital allocation and cross-country interest-rate differences. We first document a persistent geographic gradient in real interest rates: countries that are farther apart exhibit systematically larger interest-rate differentials, and the relationship is stronger for the persistent component of bilateral interest-rate gaps. We then develop a multi-country portfolio-allocation model in which geography operates through two distinct channels: bilateral investor–destination portfolio frictions and spatially correlated destination attractiveness. We estimate bilateral portfolio frictions using international portfolio holdings and identify the sensitivity of portfolio demand to returns with an instrumental-variable strategy. We recover the spatial component of destination attractiveness from its cross-country covariance structure. Holding capital stocks fixed, a decomposition attributes 43.1 percent of the geographic component of interest-rate dispersion to bilateral portfolio frictions and 56.9 percent to spatially correlated destination attractiveness. In general-equilibrium counterfactuals, removing these geographic forces substantially reduces the cross-country dispersion of capital and output, while interest rates adjust endogenously to the resulting reallocation of capital. These results show that geography matters for international finance not only through bilateral investment frictions but also through spatially correlated destination characteristics.

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1 Introduction

Geographic distance continues to shape international investment despite the rapid decline in the technological cost of moving capital across borders. Investors disproportionately hold assets issued by nearby countries, bilateral financial positions decline with distance, and cross-border portfolios remain strongly tilted toward domestic assets. These patterns are often interpreted as evidence that geography proxies for information, familiarity, or other barriers to international investment. Yet much less is known about whether these geographic forces are reflected in equilibrium returns themselves. If capital can move easily toward high-return opportunities, why should countries that are geographically close exhibit systematically more similar interest rates?

This paper studies the role of geography in generating persistent differences in interest rates and capital allocation across countries. I begin by documenting a simple empirical fact: bilateral interest-rate differentials increase with geographic distance. Doubling the distance between two countries is associated with an approximately 8-basis-point larger real interest-rate gap. More importantly, the relationship becomes stronger when I isolate the persistent component of bilateral interest-rate differentials. After removing time-varying country-specific fluctuations, doubling distance is associated with an approximately 15-basis-point larger persistent interest-rate gap. The geographic gradient therefore does not appear to be driven only by temporary regional shocks. Instead, it is particularly pronounced in the persistent component of cross-country interest-rate differences.

I interpret this pattern through two conceptually distinct ways in which geography can affect international capital allocation. The first operates "between an investor and an investment destination". Investing in geographically distant countries may involve greater information costs, lower familiarity, or other bilateral barriers, causing investors to allocate less capital to distant destinations. I refer to this channel as a bilateral portfolio friction. The second operates "across investment destinations themselves". Characteristics that make countries attractive destinations for capital may be spatially correlated: nearby countries may share institutions, financial structures, production characteristics, regional networks, or other persistent features that affect investment demand. I refer to this second channel as spatially correlated destination attractiveness.

Distinguishing these two mechanisms matters because both can generate a geographic pattern in equilibrium interest rates, but they have different economic interpretations. Suppose, for example, that a U.S. investor allocates less capital to Korea than to Canada because Korea is farther away. This is a bilateral investor–destination friction. By contrast, if Korea and Japan have similar latent characteristics that affect their attractiveness to investors because those characteristics are spatially correlated, geography operates on the destination side rather than through a particular bilateral relationship. Observing that nearby countries have similar interest rates does not by itself reveal which of these mechanisms is responsible.

To quantify their relative importance, I develop a multi-country production economy with endogenous international portfolio allocation. Households save through competitive financial intermediaries that allocate wealth across domestic and foreign destinations. The attractiveness of investing in country i from country n depends on three components: the equilibrium return in the destination, a bilateral portfolio wedge, and a destination-specific attractiveness term common to investors from all origins. Idiosyncratic investment preferences generate a logit portfolio-allocation equation. On the production side, capital is used by firms with decreasing marginal returns, so capital inflows reduce the equilibrium return to capital. International portfolio allocation and cross-country interest rates are therefore jointly determined in equilibrium.

The model allows geography to enter through both channels. The bilateral portfolio wedge increases with distance between the investor and destination. Destination attractiveness, in contrast, contains a persistent spatial component whose covariance declines with distance across destination countries. This distinction makes it possible to separate geographic segmentation arising from bilateral barriers to capital movement from geographic heterogeneity in the characteristics of investment destinations.

I estimate the bilateral portfolio friction using international portfolio holdings. The portfolio-allocation equation implies a gravity-style relationship in which bilateral distance affects investment shares after controlling for investor-by-time and destination-by-time factors. The estimated distance elasticity is large: a 1 percent increase in bilateral distance is associated with an approximately 0.85 percent reduction in bilateral portfolio holdings. Doubling distance therefore reduces predicted bilateral holdings by roughly 45 percent. The estimates also reveal substantial home bias in international portfolio allocation.

A separate identification problem arises because equilibrium returns are endogenous to capital allocation. To identify the sensitivity of portfolio demand to returns, I use an instrumental-variable strategy based on residual asset demand, adapted from Koijen and Yogo (2026). I construct predicted demand for each destination using the estimated gravity component of portfolio allocation and combine it with predicted asset supply based on destination characteristics. The instrument uses demand originating from countries other than the investor whose portfolio choice is being studied, thereby excluding the bilateral investor–destination component that enters the portfolio equation directly. The resulting variation in relative demand pressure shifts equilibrium returns through market clearing and identifies the return sensitivity of portfolio demand. The estimates imply a taste-dispersion parameter of approximately $\sigma_N = 0.04$.

I then turn to the second geographic channel. After recovering destination attractiveness, I decompose its residual component into three parts: a spatially correlated persistent component, a non-spatial persistent country component, and transitory variation. The spatial covariance structure is parameterized so that correlation decays exponentially with bilateral distance. Cross-country covariance identifies the spatial component, while within-country covariance over time separates persistent non-spatial heterogeneity from transitory noise.

The estimated spatial component is economically important. It accounts for approximately 56 percent of the residual variation in destination attractiveness and about 61 percent of persistent cross-country heterogeneity. The estimated spatial decay parameter is 2,842 kilometers, implying a covariance half-life of approximately 1,970 kilometers. A bootstrap likelihood-ratio test rejects the absence of a spatial persistent component at conventional significance levels. I recover country-specific spatial components using a best linear unbiased predictor based on the estimated covariance structure.

With the two geographic channels separately identified, I quantify their contribution to cross-country interest-rate dispersion. I first conduct a partial-equilibrium decomposition that holds capital stocks fixed. Under the baseline, the average pairwise interest-rate dispersion is 1.80 percentage points. Removing bilateral distance frictions reduces dispersion to 1.59 percentage points, while removing the spatial component reduces it to 1.35 percentage points. Removing both lowers dispersion further to 0.87 percentage points. A Shapley decomposition attributes 43.1 percent of the geographic component of interest-rate dispersion to bilateral portfolio frictions and 56.9 percent to spatially correlated destination attractiveness.

The general-equilibrium results reveal an important distinction between the direct effect of geography on interest-rate differences and its broader effect on capital allocation. When capital stocks are allowed to adjust endogenously, removing geographic forces substantially compresses the cross-country distribution of capital and output. Eliminating bilateral distance frictions reduces the standard deviation of log capital by 19.6 percent, while eliminating both geographic channels reduces it by 43.3 percent. Output dispersion also falls. Interest-rate dispersion, however, need not move in the same direction. For example, removing bilateral distance frictions increases interest-rate dispersion in general equilibrium even though it reduces dispersion when capital stocks are held fixed. As capital reallocates across countries, marginal products adjust endogenously and can offset—or even reverse—the direct effect of removing geographic frictions on observed returns.

This result changes how the geographic gradient in interest rates should be interpreted. The central implication is not simply that geographic frictions mechanically widen interest-rate differentials. Rather, geography creates persistent segmentation in the allocation of global capital. Interest-rate differences are an equilibrium manifestation of this segmentation, but endogenous capital adjustment can mask substantial changes in the underlying allocation. Consequently, the quantitative importance of geography is more clearly visible in the distribution of capital and output than in interest-rate dispersion alone.

The paper contributes to several strands of literature. First, it contributes to the literature on international portfolio allocation and financial gravity, which documents that bilateral asset holdings decline with geographic distance and emphasizes information, familiarity, and related barriers to foreign investment. I complement this literature by linking geographic portfolio frictions to equilibrium interest-rate differences and by distinguishing bilateral frictions from spatial heterogeneity on the destination side.

Second, the paper contributes to the literature on segmented international financial markets. Building on frameworks in which limited international risk sharing and intermediary frictions affect asset prices and capital flows, the model introduces an explicit spatial structure into international portfolio allocation. The results show that segmentation has both a bilateral component, arising from investor–destination frictions, and a destination-side component, arising from spatially correlated characteristics of investment opportunities.

Third, the paper provides a quantitative decomposition of the role of geography in international capital allocation. Rather than treating distance as a single reduced-form barrier, I separate two channels that are observationally similar in bilateral interest-rate data but economically distinct. The decomposition shows that neither channel alone accounts for the observed geographic pattern. Spatially correlated destination characteristics are at least as important quantitatively as bilateral portfolio frictions.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 documents the geographic gradient in bilateral interest-rate differentials and its persistent component. Section 4 develops the model. Section 5 estimates bilateral portfolio frictions and the sensitivity of portfolio allocation to returns. Section 6 estimates and recovers the spatial component of destination attractiveness. Section 7 presents the partial-equilibrium decomposition and general-equilibrium counterfactuals. Section 8 concludes.

Related Literatures

2 Data

3 Motivating Empirical Evidence

This section investigates the spatial relationship in interest rates across countries. If frictions to international capital allocation have a geographic dimension, or if factors relevant for interest rates are spatially correlated, interest rates should exhibit a systematic geographic pattern across countries. In particular, countries that are geographically closer may display more similar interest rates, whereas interest-rate differentials may increase with bilateral distance. To examine this relationship, I consider two empirical specifications. The first exploits time variation in bilateral interest-rate differentials, while the second isolates their time-invariant, persistent component.

3.1 Geographic Gradient in Interest-Rate Differentials

We consider following regression.

$$r_{high,t} - r_{low,t} = \beta \ln(dist_{high,low}) + X_{high,low}\Gamma + \alpha_{\text{high}} + \alpha_{\text{low}} + \varepsilon_{ijt}$$

For each country pair, the interest-rate differential is defined as the interest rate of the high-interest-rate country minus that of the low-interest-rate country. Interest rates are measured in percentage points, so that a one-unit difference in the dependent variable corresponds to a one-percentage-point interest-rate gap. The vector $X_{\text{high,low}}$ contains bilateral controls, including indicators for whether the two countries share a border or a common language, have a colonial relationship, or were historically part of the same country. The specification also includes fixed effects for the high- and low-interest-rate countries, α_{high} and α_{low}, respectively, as well as year fixed effects λ_t. The coefficient of interest, β, measures how the interest-rate gap varies with geographic distance. A positive value of β indicates that the interest-rate differential tends to widen as the geographic distance between the two countries increases. The high- and low-country fixed effects absorb systematic differences associated with each country depending on its position in the interest-rate ranking. Intuitively, the country fixed effects allow us to compare how the interest-rate gap varies across partners of a given country. For example, conditional on the fixed effect associated with the United States, the regression compares whether the U.S. interest-rate gap tends to be larger with geographically more distant countries.

Table 1: Geographic Distance and Interest-Rate Differentials

	(1)	(2)	(3)	(4)
Log distance	0.115***	0.108**	0.121***	0.117***
	(0.036)	(0.044)	(0.034)	(0.043)
Shares border		-0.046		-0.096
		(0.094)		(0.098)
Common language			0.076	0.063
			(0.078)	(0.086)
Colonial relationship				0.109
				(0.078)
Same country (past/now)				0.105
				(0.096)
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	25501	25501	25501	25501
Within R-squared	0.000	0.000	0.000	0.000

Table 1 shows that geographically more distant country pairs tend to exhibit larger interest-rate differentials. An estimate of $\beta = 0.115$ implies that doubling bilateral distance is associated with an approximately 8-basis-point increase in the interest-rate gap. The coefficient remains similar in magnitude after controlling for bilateral relationships, including shared borders, common language, colonial ties, and historical state relationships, suggesting that the geographic gradient is not driven by these bilateral characteristics alone.

3.2 The Persistent Component of Interest-Rate Differentials

The baseline specification establishes that interest-rate differentials are larger between geographically distant countries. However, this relationship may partly reflect transitory country-specific shocks. For example, geographically proximate countries may experience similar monetary or macroeconomic shocks in a given year, generating temporarily similar interest rates. I therefore examine whether geographic distance is also related to the persistent component of bilateral interest-rate differentials.

I first decompose the bilateral interest-rate differential into high-country-by-year and low-country-by-year components and a time-invariant country-pair component, $\delta_{\text{high,low}}$. The country-by-year fixed effects absorb time-varying shocks specific to either member of the pair. The estimated pair component therefore captures the portion of the bilateral interest-rate differential that persists over time after removing these country-specific fluctuations.

$$r_{high,t} - r_{low,t} = \alpha_{high,t} + \alpha_{low,t} + \underbrace{\delta_{high,low}}_{\text{pair FE}} + \lambda_t + u_{high,low,t}$$

In the second stage, I regress the estimated pair component on bilateral geographic distance and the same set of bilateral controls used in the baseline specification.

$$\hat{\delta}_{high,low} = \beta^P \ln(dist_{high,low}) + X_{high,low}\Gamma + \eta_{high,low}$$

Table 2 reports the regression estimates, while Figure 1 presents the corresponding binscatter of the persistent interest-rate differential against log bilateral distance. Both show a strong positive geographic gradient. The baseline estimate of $\beta^P = 0.211$ implies that doubling bilateral distance is associated with an approximately 15-basis-point larger persistent interest-rate gap. The coefficient remains between 0.21 and 0.23 across specifications that control for bilateral relationships.

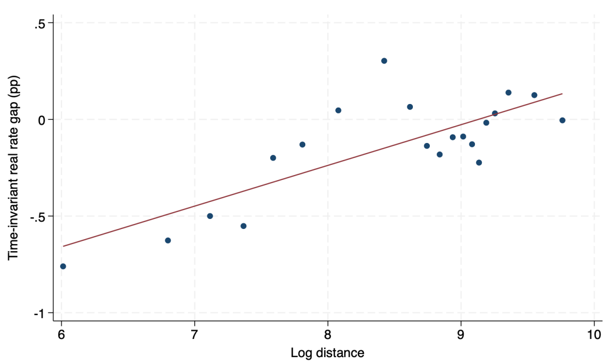
The estimated geographic gradient is numerically larger than in the pooled specification, where doubling distance is associated with an approximately 8-basis-point larger interest-rate differential.

This pattern suggests that the relationship between geographic distance and interest-rate dispersion is not driven solely by transitory country-level shocks, but is particularly pronounced in the persistent component of bilateral interest-rate differences.

Table 2: Geographic Distance and the Persistent Component of Interest-Rate Differentials

	(1)	(2)	(3)	(4)
Log distance	0.211***	0.224***	0.211***	0.230***
	(0.014)	(0.015)	(0.014)	(0.016)
Shares border		0.175**		0.122
		(0.082)		(0.087)
Common language			-0.001	-0.029
			(0.047)	(0.048)
Colonial relationship				0.094
				(0.090)
Same country (past/now)				0.230*
				(0.131)
Observations	3060	3060	3060	3060
R-squared	0.069	0.071	0.069	0.072

Figure 1: Persistent Interest-Rate Differentials and Geographic Distance



4 Model

In this section, we develop a dynamic spatial production economy with multiple countries. There are N countries, indexed by $n \in \{1, \ldots, N\}$. Each country is populated by a representative household and competitive firms. Households choose consumption and saving, while their savings are allocated across domestic and foreign investment opportunities through financial intermediaries. International portfolio allocation is subject to bilateral frictions, so that otherwise identical investment opportunities may be valued differently across investor–destination pairs. This setup is consistent with the recent literature on segmented international capital markets, following Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021, 2025), and is closely related to the international portfolio-allocation frameworks in Pellegrino et al. (2025) and Kleinman et.al. (2026).

4.1 Households

The representative household in country n is endowed with a fixed amount of labor, ℓ_n. The household chooses consumption and saving to maximize lifetime utility,

$$\sum_{t=0}^{\infty} \beta_n^t \, u(c_{nt}) \tag{1}$$

$$\text{subject to } c_{n,t} + a_{n,t+1} = \mathcal{R}_{n,t} a_{n,t} + w_{n,t} \ell_n$$

where β_n is the country-specific discount factor,[1] and $\mathcal{R}_{n,t}$ denotes the return on the household's savings portfolio. The optimal consumption-saving decision satisfies $u'(c_{n,t}) = \beta_n \mathcal{R}_{n,t+1} u'(c_{n,t+1})$. In steady state, consumption and portfolio returns are constant over time, so the Euler equation reduces to $1 = \beta_n \mathcal{R}_n$.

4.2 International Capital Allocation

An investor in country n considering investment in country i faces a bilateral portfolio wedge Φ_{ni}. The wedge captures frictions that reduce the relative attractiveness of investing in a particular foreign destination. We assume that the wedge increases with bilateral geographic distance. One interpretation is that geographic distance is associated with greater informational frictions or lower familiarity with foreign investment opportunities.

The term ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. Unlike the bilateral portfolio wedge Φ_{ni}, which varies across investor–destination pairs, ω_{it} depends only on the destination country i and time t. It therefore captures characteristics

[1] Cross-country differences in discount factors may reflect differences in mortality risk, as in Pellegrino et al. (2025), or differences in time preferences, as documented by Falk et al. (2018).

of country i that affect its attractiveness as an investment destination uniformly across investors, beyond the observed interest rate r_{it}. A higher ω_{it} increases the portfolio share allocated to country i by investors from all countries. This distinction is important for the analysis below. While Φ_{ni} captures bilateral segmentation between investor n and destination i, ω_{it} captures destination-side heterogeneity that is common across investors.

For each unit of savings, the intermediary draws an idiosyncratic taste shock ε_{ni} for each destination i, independently across destinations, from a Gumbel distribution.

$$F(x) = exp(-exp(-x + \bar{\gamma}))$$

where $\bar{\gamma}$ is Euler–Mascheroni constant. The scale of the idiosyncratic shock is governed by σ_N. The intermediary in country n invests in country i if the friction- and preference-adjusted return from investing in i is the highest among all available destinations.

$$\varepsilon(n, i, \nu)\sigma_N - \Phi_{ni} + \omega_{it} + r_{i,t} = \max_h \{\varepsilon(n, i, \nu)\sigma_N - \Phi_{nh} + \omega_{ht} + r_{h,t}\} \tag{2}$$

The share of country n's savings allocated to country i through financial intermediaries is given by

$$b_{nit} = \frac{a_{nit}}{a_{nt}} = \frac{\exp(-\Phi_{ni} + \omega_i + r_{i,t})^{1/\sigma_N}}{\sum_h \exp(-\Phi_{nh} + \omega_h + r_{h,t})^{1/\sigma_N}} \tag{3}$$

Here, a_{nit} denotes the amount of country n's savings allocated to country i, while a_{nt} denotes total savings in country n. A larger portfolio wedge Φ_{ni} reduces the share allocated to destination i, whereas a higher destination-specific attractiveness ω_{it} or a higher interest rate r_{it} increases its portfolio share. The parameter σ_N governs the dispersion of idiosyncratic investment preferences and, therefore, the sensitivity of portfolio allocation to differences in returns and destination characteristics.

The gross return on the household's savings portfolio is then the portfolio-share-weighted average of destination returns:

$$\mathcal{R}_{n,t} = 1 + \sum_i b_{nit} r_{i,t} \tag{4}$$

4.3 Firm

Each country n is populated by a representative firm operating a Cobb–Douglas production technology. The firm takes the wage $w_{n,t}$ and the interest rate $r_{n,t}$ as given and chooses its capital stock to maximize the present value of profits. Output is produced using capital $k_{n,t}$ and labor $\ell_{n,t}$, with country-specific productivity $z_{n,t}$. Capital depreciates at rate δ, so that investment required to carry capital into the next period is $k_{n,t+1} - (1-\delta)k_{n,t}$.

The optimality condition for capital implies that the equilibrium return to capital is equal to its marginal product net of depreciation,

$$r_{n,t+1} = \alpha z_{n,t} k_{n,t}^{\alpha-1} \ell_{n,t}^{1-\alpha} - \delta.$$

Thus, the interest rate in each country is decreasing in the amount of capital allocated to that country. An increase in capital inflows raises $k_{n,t}$, lowers the marginal product of capital, and thereby reduces the equilibrium interest rate. This relationship provides the link between international portfolio allocation and cross-country interest-rate differences in the model.

4.4 Market Equilibrium

A competitive equilibrium consists of allocations $\{\{a_{nt}\}_{n=1}^N,\ \{b_{nit}\}_{n,i=1}^N,\ \{k_{nt}\}_{n=1}^N\}$ and prices $\{\{\mathcal{R}_{nt}\}_{n=1}^N,\ \{r_{nt}\}_{n=1}^N\}$ such that, for all n and t:

1. Agents solve the optimization problem in (1), taking $\{\mathcal{R}_{nt}\}$ and $\{r_{nt}\}$ as given.

2. The rental rate equals the net marginal product of capital:

$$r_{nt} = \alpha z_n k_{nt}^{\alpha-1} - \delta. \tag{5}$$

3. The capital market clears in each location:

$$k_{nt} = \sum_i b_{int}\, a_{it}. \tag{6}$$

4. The gross return on a unit of wealth is

$$\mathcal{R}_{nt} = 1 + \sum_i b_{nit}\, r_{it}. \tag{7}$$

5 Parameterization

Table 3 details model parameters.

Table 3: Parameterization

Symbol	Value	Description	Source / Method
I. Estimated Parameters			
Φ_{ij}	—	Portfolio wedge	Gravity regression
$\omega_{i,t}$	—	Country attractiveness	PPML Estimation
σ_N	0.04	Taste dispersion	IV regression
II. Calibration			
β_n	—	Discount factor	$\beta_n \mathcal{R}_n = 1$
$z_{n,t}$	—	TFP	Target $r_{n,t}$

5.1 Portfolio Friction Estimation

This section estimates bilateral portfolio frictions, $\phi_{ni} = \Phi_{ni}/\sigma_N$, using PPML. From the portfolio allocation equation (3), investment shares can be decomposed into three components. First, the bilateral portfolio friction term ϕ_{ni} varies across investor–destination pairs and captures barriers specific to capital allocation between countries. I parameterize this component as a function of bilateral geographic distance. Specifically we assume $\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta^{foreign} \mathbb{1}[n = i]$. Second, the destination-specific component, $(\omega_{it} + r_{it})/\sigma_N$, is common across investors and is absorbed by destination-country $\times$ time fixed effects. Third, the denominator of the allocation equation captures investor-specific aggregate demand for foreign assets and is absorbed by investor-country $\times$ time fixed effects.

$$b_{nit} = \frac{\overbrace{\exp(-\phi_{ni})}^{\substack{\text{Capital Moving}\\ \text{Friction}}} \overbrace{\exp(\omega_{it}/\sigma_N + r_{it}/\sigma_N)}^{\substack{\text{Time}\times\\ \text{Destin FE}}}}{\underbrace{\sum_h \exp(-\phi_{nh} + \omega_{ht}/\sigma_N + r_{ht}/\sigma_N))}_{\text{Time}\times\text{Origin FE}}} \tag{8}$$

Estimation Results Table 4 reports the PPML estimates of bilateral portfolio frictions. The coefficient on log distance is negative and statistically significant, indicating that geographic distance reduces bilateral portfolio allocation. The estimate of $\beta_d = -0.854$ implies that a 1 percent increase in bilateral distance is associated with a 0.854 percent decline in bilateral portfolio holdings. In terms of economic magnitude, doubling bilateral distance is associated with an approximately 45 percent reduction in predicted bilateral portfolio holdings.

The large negative coefficient on the foreign indicator captures substantial home bias in international portfolio allocation. These results imply that geographic distance represents an important bilateral portfolio friction in international capital allocation.

Table 4: Bilateral Portfolio Frictions and Geographic Distance

Table 5: PPML Estimation Results

	(1)
$\ln(dist_{i,j})$	-0.854***
	(0.0757)
Foreign	-3.921***
	(0.158)
Observations	33506

Standard errors in parentheses

Standard errors clustered by iso_o and iso_d

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2 σ_N Estimation

The return term is endogenous because equilibrium interest rates are jointly determined by capital demand and capital allocation. To estimate σ_N, we use IV approach adapted from Koijen and Yogo (2026). The intuition is that changes in the supply of investable assets shift equilibrium returns through market clearing, but do not directly affect investors' portfolio preferences conditional on destination and investor fixed effects. Therefore, residual asset supply provides exogenous variation in returns that identifies the sensitivity of portfolio allocation to expected returns. Equation (3) can be modified to be expressed as

$$\log \frac{b_{ni}}{b_{nn}} = \frac{1}{\sigma_N}(r_{it} - r_{nt} - \Phi_{ni} + \omega_{it} - \omega_{nt})$$

The identification assumptions for the IV strategy are:

$$\begin{aligned} \text{Exogeneity:} \quad & \text{Cov}(\varepsilon_{nit}, dist_{ji} \mid dist_{ni}, \iota_i, \eta_n) = 0, \\ \text{Relevance:} \quad & \text{Cov}(r_{it}, dist_{ji} \mid dist_{ni}, \iota_i, \eta_n) \neq 0, \qquad \forall i \neq j. \end{aligned}$$

Demand Component From equation (3), we construct the time-invariant component of portfolio demand based only on estimated bilateral portfolio frictions. Specifically, the predicted allocation share from investor country m to destination country i is given by

$$\hat{b}_{mit}^{gravity} = \frac{\exp(-\phi_{mi})}{\sum_h \exp(-\phi_{mh})}.$$

This component captures the geographic pattern of international portfolio allocation implied by bilateral frictions, abstracting from time-varying destination characteristics and equilibrium returns. We then aggregate the predicted demand from all foreign investors, excluding the demand originating from country n, as

$$\sum_{m \neq n} A_{mt} \hat{b}_{mit}^{gravity}.$$

Supply Component To construct the supply component, we estimate the following panel regression:

$$\log S_{it} = \Psi' z_t(i)$$

where

$$z_t(i) = [\log GDP_t(i), \log pop_t(i)]'.$$

The predicted value captures the component of asset supply explained by observable country characteristics: $\log \hat{S}_{it} = \hat{\Psi}' z_t(i)$.

Residual Supply Using the predicted asset allocation from the gravity model, we construct the residual supply component:

$$Z_{nit} = \log \left(\sum_{m \neq n} A_{mt} \hat{b}_{mit}^{gravity} \right) - \log \hat{S}_{it}$$

For interpretation, we take the negative of the conventional residual supply measure such that a larger value of Z_{nit} represents stronger demand pressure for destination country i. Intuitively, Z_{nit} captures the extent to which asset demand for country i exceeds the level predicted by its observable supply characteristics. A country experiencing stronger-than-expected demand pressure receives larger capital inflows, which affects its equilibrium return through the capital market clearing condition.

The identification assumption is that the residual demand pressure affects portfolio allocation only through its effect on the equilibrium return of destination country i. The leave-one-out construction excludes the bilateral component associated with investor country n, ensuring that the instrument is not mechanically correlated with the bilateral portfolio friction between n and i.

2SLS The return elasticity of portfolio demand is estimated using a two-stage least squares procedure. The first stage is

$$r_{it} - r_{nt} = \pi Z_{nit}^{rel} + \mathbf{X}\beta + \nu_{nit},$$

where

$$Z_{nit}^{rel} = Z_{-n,t}^{i} - Z_{-n,t}^{n}$$

denotes the relative residual supply component between destination country i and investor country n. The second stage estimates the sensitivity of portfolio allocation to returns:

$$\log \frac{b_{nit}}{b_{nnt}} = \lambda_r(\widehat{r_{it} - r_{nt}}) + \beta_d \log dist_{ni} + \beta^f Foreign_{ni} + \iota_i + \eta_n + \varepsilon_{nit}.$$

The estimated return coefficient implies $\hat{\sigma}_N = \frac{1}{\hat{\lambda}_r} \approx 0.04$

Table 6: IV Estimation Results

	(1)	(2)
	First stage	2SLS
Relative Z	0.01***	
	(0.00)	
ΔR		25.21***
		(6.10)
log(distance)	-0.00	-1.14***
	(0.00)	(0.02)
First-stage F (excl. instr.)	186.08	186.08
Kleibergen-Paap rk Wald F		186.08
Observations	28852	28852
Adjusted R^2		0.022

Fixed effects: country and counterpart. Robust standard errors in parentheses.

6 Spatial Structure in ω

6.1 Motivation and Decomposition

In the model, ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. If destination characteristics are spatially correlated, geographically proximate countries should exhibit more similar values of ω_{it}. This provides a second channel through which geography may affect international capital allocation, distinct from the bilateral portfolio friction ϕ_{ni}.

$$\omega_{high,t} - \omega_{low,t} = \beta \ln(dist_{high,low}) + X_{high,low}\Gamma + \alpha_{high} + \alpha_{low} + \alpha_t + \varepsilon_{ijt}$$

Table 7: Distance Regression on ω

	(1)	(2)	(3)	(4)
ln(distance)	0.0014***	0.0014***	0.0015***	0.0015***
	(0.0004)	(0.0005)	(0.0004)	(0.0005)
Contiguity		0.0001		0.0010
		(0.0011)		(0.0013)
Common language			0.0008	0.0015
			(0.0014)	(0.0014)
Colonial link				-0.0032*
				(0.0017)
Same country				-0.0056**
				(0.0028)
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	14,817	14,817	14,817	14,817
Country pairs	1,203	1,203	1,203	1,203
Within R^2	0.000	0.000	0.000	0.001

We decompose

$$\omega_{it} = X_{it}'\Gamma + \alpha_t + s_i + a_i + \varepsilon_{it}$$

$$s \sim N(0, \Sigma_s), \qquad a \sim N(0, \sigma_a^2 I), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2),$$

$$[\Sigma_s]_{ij} = \sigma_s^2 \exp\left(-\frac{d_{ij}}{\rho}\right)$$

Here, $X_{it}'\Gamma$ captures the component explained by observable destination characteristics. The remaining variation is decomposed into two persistent country-level components and a transitory component. The term s_i denotes the spatially correlated persistent component, so that countries located closer to one another tend to have more similar values of s_i. In contrast, a_i captures persistent country-specific heterogeneity that is not spatially correlated across countries. Finally, ε_{it} represents transitory variation in destination attractiveness. Both s_i and a_i are time invariant, allowing persistent cross-country heterogeneity to be separated into spatial and non-spatial components.

The model therefore features two conceptually distinct geographic channels. The first is bilateral geography, captured by ϕ_{ni}, which depends on the distance between investor country n and destination country i. For example, from the perspective of a U.S. investor, Canada is geographically closer than Korea and therefore faces a smaller distance-related portfolio friction. The second is destination-side geography, captured by s_i, which reflects spatial correlation in destination attractiveness across countries. For example, Korea is geographically closer to Japan than to France, so the spatial component of Korea's destination attractiveness is more strongly related to that of Japan than to that of France. Thus, geography affects international capital allocation through both bilateral portfolio frictions and spatially correlated destination characteristics.

6.2 Identification

To identify the spatial component, we first remove observable destination characteristics and common time effects from destination attractiveness. Define the residual as

$$u_{it} = \omega_{it} - X_{it}'\hat{\Gamma} - \hat{\alpha}_t = s_i + a_i + \varepsilon_{it}.$$

We assume that the spatial component, the non-spatial persistent component, and the transitory component are mutually orthogonal. We also assume that the non-spatial persistent component is uncorrelated across countries and that the transitory component is uncorrelated across countries and over time. Identification comes from comparing covariance across countries and across time. For two different countries $i \neq j$,

$$\mathrm{Cov}(u_{it}, u_{js}) = \sigma_s^2 \exp\left(-\frac{d_{ij}}{\rho}\right).$$

Because a_i is country-specific and non-spatial and ε_{it} is transitory, cross-country covariance reflects only the spatial component. The cross-country covariance–distance relationship jointly identifies σ_s^2 and ρ: σ_s^2 governs the scale of spatial covariance, while ρ governs the rate at which covariance decays with distance. For the same country observed in two different periods,

$$\mathrm{Cov}(u_{it}, u_{is}) = \sigma_s^2 + \sigma_a^2, \qquad t \neq s.$$

Since σ_s^2 is already identified from cross-country covariance, the remaining persistent variation identifies σ_a^2. Finally,

$$\mathrm{Var}(u_{it}) = \sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2.$$

Thus, the remaining within-country variation identifies the transitory variance σ_ε^2.

6.3 Estimates

We jointly estimate the variance parameters $(\sigma_s^2, \rho, \sigma_a^2, \sigma_\varepsilon^2)$ by restricted maximum likelihood (REML). Table 8 reports the estimates.

The estimated variance of the spatial persistent component is $\hat{\sigma}_s^2 = 0.00668$, accounting for 55.7 percent of the total residual variation in destination attractiveness. The non-spatial persistent component has variance $\hat{\sigma}_a^2 = 0.00428$, accounting for 35.7 percent, while the transitory component accounts for the remaining 8.6 percent. Among the two persistent components alone, spatial variation accounts for approximately 61 percent of cross-country heterogeneity.

The estimated spatial decay parameter is $\hat{\rho} = 2{,}842$ km. Under the exponential covariance specification, spatial covariance falls to one half of its initial level at a distance of $\rho \ln 2$, which corresponds to approximately 1,970 km. Thus, destination attractiveness exhibits substantial spatial dependence over economically meaningful distances.

Finally, a parametric-bootstrap likelihood-ratio test of the null hypothesis $H_0 : \sigma_s^2 = 0$ yields an LR statistic of 5.01 and a bootstrap p-value of 0.040, providing evidence against the absence of a spatial persistent component.

Table 8: Spatial Decomposition of ω

Parameter	Estimate	Share of Var.	Interpretation
σ_s^2	0.00668	55.7%	Spatial persistent
ρ	2,842 km	—	Spatial decay parameter
σ_a^2	0.00428	35.7%	Non-spatial persistent
σ_ε^2	0.00103	8.6%	Transitory
Total	0.01199	100%	

6.4 Recovering the Spatial Component

After estimating the variance components, we recover the country-specific spatial and non-spatial persistent components using the Best Linear Unbiased Predictor (BLUP). This approach is closely related to the empirical Bayes and BLUP methods used in the labor economics literature to recover latent persistent heterogeneity from noisy observed outcomes, as in Kane and Staiger (2008) and Chetty et al. (2014). We apply the same basic idea here, using the estimated spatial covariance structure to separate the latent spatial component from non-spatial country-specific heterogeneity and transitory noise. We first average the residualized destination attractiveness within each country over time:

$$\bar{u}_i = \frac{1}{T_i} \sum_t u_{it} = s_i + a_i + \bar{\varepsilon}_i,$$

where T_i is the number of observations for country i. In vector form,

$$\bar{u} = s + a + \bar{\varepsilon}.$$

Because averaging reduces the variance of the transitory component, the covariance matrix of $\bar{u}$ is

$$V_{\bar{u}} = \Sigma_s + \sigma_a^2 I + D_\varepsilon,$$

where

$$D_\varepsilon = \operatorname{diag}\left(\frac{\sigma_\varepsilon^2}{T_1}, \ldots, \frac{\sigma_\varepsilon^2}{T_N}\right).$$

The spatial component is recovered from its conditional expectation given the observed average residuals:

$$\hat{s} = E[s \mid \bar{u}] = \Sigma_s V_{\bar{u}}^{-1} \bar{u}.$$

Similarly, the non-spatial persistent component is recovered as

$$\hat{a} = E[a \mid \bar{u}] = \sigma_a^2 I V_{\bar{u}}^{-1} \bar{u}.$$

The predictor assigns variation in $\bar{u}$ to the spatial component according to the estimated cross-country covariance structure. Variation that is persistent but not explained by spatial covariance is instead attributed to the non-spatial country component.

7 Quantitative Exercise

7.1 Partial-Equilibrium Decomposition of Geographic Channels

We first quantify the direct contribution of the two geographic channels to cross-country interest-rate dispersion. In this exercise, we hold country-level capital stocks fixed at their baseline values and recompute implied interest rates under alternative configurations of bilateral portfolio frictions and spatial destination attractiveness. Holding capital stocks fixed isolates the direct contribution of each geographic channel from the general-equilibrium adjustment of capital across countries.

$$D(r) = \frac{2}{N(N-1)} \sum_{i<j} |r_i - r_j| \tag{9}$$

Table 9: Counterfactual scenarios

	Spatial component s	
Distance friction ϕ	Present	Removed
Present	r^{11}	r^{10}
Removed	r^{01}	r^{00}

The baseline dispersion is 1.80 percent. Removing bilateral distance frictions reduces dispersion to 1.59 percent, while removing the spatial component of destination attractiveness reduces it to 1.35 percent. Removing both geographic channels reduces dispersion further to 0.87 percent. Because the effects of the two channels interact, we use a Shapley decomposition that averages the marginal contribution of each channel across the two possible states of the other channel. The decomposition attributes 43.1 percent of the geographic component to bilateral portfolio frictions and 56.9 percent to spatially correlated destination attractiveness.

Project/Figures/partial_decomposition.png

7.2 General-Equilibrium Counterfactuals

We next allow capital stocks to adjust endogenously and solve for the steady-state equilibrium under the same four scenarios. Unlike the decomposition above, this exercise incorporates the general-equilibrium response of capital allocation, output, and interest rates to changes in geographic frictions.

Table 10 reports the general-equilibrium counterfactuals. Removing bilateral distance frictions substantially reduces the dispersion of capital across countries: the standard deviation of log capital falls by 19.6 percent relative to the baseline. Output dispersion also falls by 7.9 percent. However, interest-rate dispersion increases from 1.59 percent to 2.07 percent. This contrasts with the partial-equilibrium decomposition above, where removing distance frictions directly compresses interest-rate differences. The difference reflects the endogenous reallocation of capital in general equilibrium. Removing the spatial destination component reduces interest-rate dispersion from 1.59 percent to 1.32 percent and lowers the dispersion of capital and output by 9.1 percent and 3.3 percent, respectively. When both geographic channels are removed, capital dispersion falls by 43.3 percent and output dispersion by 13.4 percent. Interest-rate dispersion, however, remains close to its baseline level at 1.57 percent, highlighting that the general-equilibrium adjustment of capital can substantially offset the direct effect of geographic frictions on interest-rate dispersion.

Table 10: Counterfactual by Scenario

	$D(r)$ (%)			STD(log k)		STD(log y)	
	All	AE	EME	Level	Δ vs. baseline (%)	Level	Δ vs. baseline (%)
Baseline	1.59	0.53	1.73	1.486	–	1.382	–
No distance friction	2.07	0.83	2.32	1.195	−19.6	1.273	−7.9
No spatial component	1.32	0.54	1.42	1.351	−9.1	1.336	−3.3
Neither channel	1.57	1.10	1.70	0.843	−43.3	1.196	−13.4

8 Conclusion

This paper studies how geography shapes international capital allocation and cross-country interest-rate differences. We first document a robust spatial gradient in interest rates: geographically distant countries exhibit larger interest-rate differentials, and this relationship is particularly pronounced in the persistent component of interest-rate gaps. These patterns suggest that geography remains relevant for international financial allocation even beyond short-run country-specific fluctuations.

To quantify the mechanisms behind this relationship, we develop a multi-country portfolio-allocation model with two distinct geographic channels. The first is a bilateral portfolio friction that depends on the relationship between an investor country and an investment destination. The second operates through destination-specific attractiveness, which may itself be spatially correlated across countries. We estimate bilateral portfolio frictions from international portfolio holdings. We then decompose destination attractiveness into spatially correlated persistent, non-spatial persistent, and transitory components. The estimated spatial component accounts for a substantial share of persistent cross-country heterogeneity in destination attractiveness.

Holding capital stocks fixed, the decomposition shows that both geographic channels contribute meaningfully to cross-country interest-rate dispersion. Bilateral portfolio frictions account for 43.1 percent of the geographic component, while spatially correlated destination attractiveness accounts for 56.9 percent. When capital stocks are allowed to adjust in the general-equilibrium counterfactuals, removing geographic frictions substantially reduces the dispersion of capital and output across countries, while the response of interest-rate dispersion reflects the endogenous reallocation of capital.

Overall, the results show that the role of geography in international finance extends beyond bilateral barriers to investment. Spatial correlation in destination characteristics constitutes an additional and quantitatively important channel through which geography shapes global capital allocation. Distinguishing these two channels is therefore important for understanding persistent differences in capital allocation and returns across countries.

References